

Step 1: S_{Osm}

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Step 2: U_{Osm}

Step 3: U_{Na}

Cases

Step 1: S_{Osm}

Step 2: U_{Osm}

Step 3: U_{Na}

Cases

Serum Osm
Is it real?

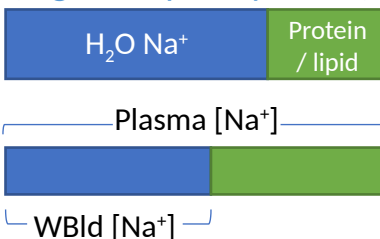
Na^+

>290

<290

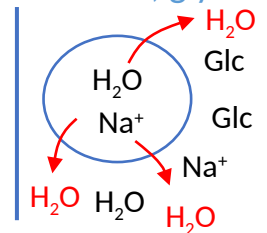
Artifact

↑ TGs or ↑ protein
(e.g. multiple myeloma)



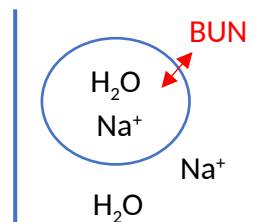
Active Osm

↑ glucose, mannitol,
sorbitol, glycerin



Inactive Osm

BUN, ETOH



Hyponatremia associated with ↑ or normal serum osmolarity (Osm) is generally not “true hyponatremia.”

Hypo-osmolar hyponatremia

Pseudohyponatremia (Artifact)

Our standard plasma Na measures the [Na] in whole plasma. When there are ↑ lipids (i.e. hypertriglyceridemia) or ↑ protein (i.e., multiple myeloma) this will dilute the whole plasma to give a falsely low [Na], even though the [Na] in the water plasma is still normal. Send a ‘whole blood’ [Na] to confirm the true [Na].

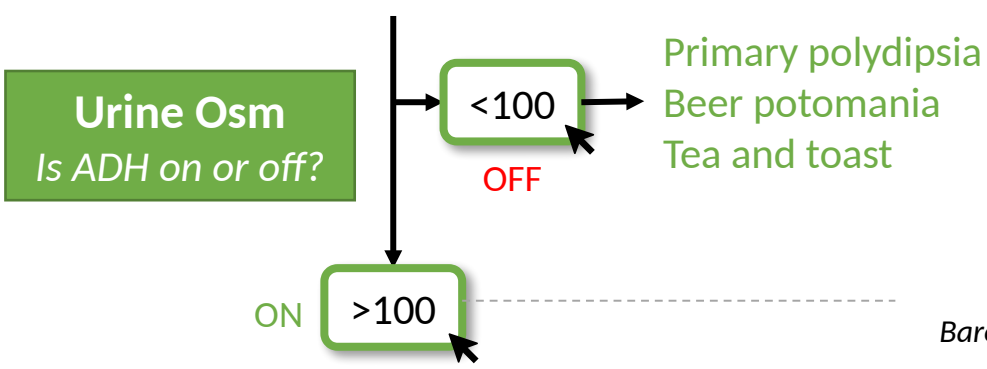
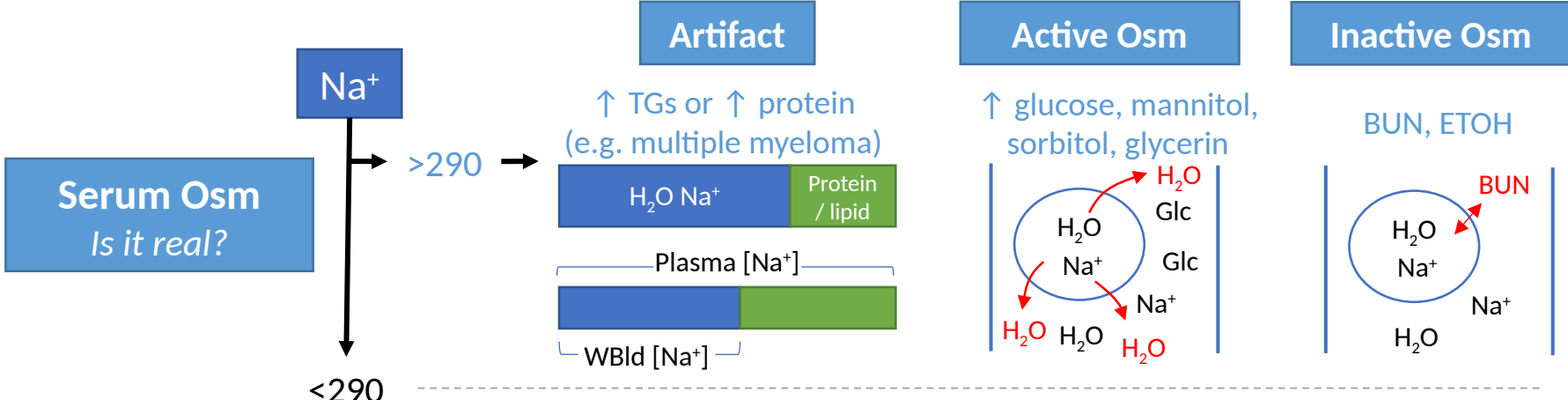
Active Osm

Some compounds (most commonly glucose) do not freely cross the cell wall and osmotically pull water out of the cell, into plasma in order to equilibrate the Osm gradient. This consequently dilutes the concentration of Na.

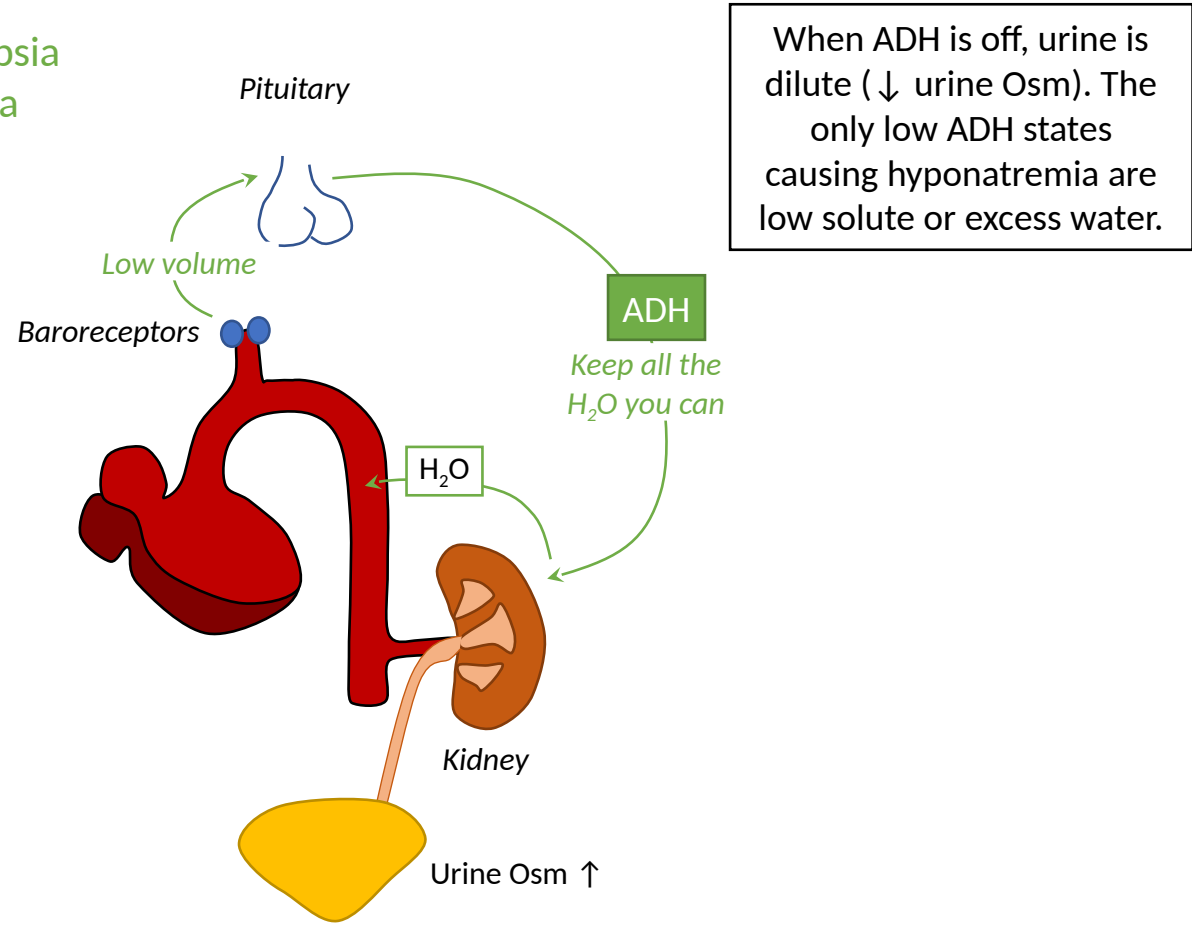
Inactive Osm

Some compounds (such as BUN, ethanol) freely cross the cell membrane and thus do not pull water into plasma. However, these Osm are often 2/2 another primary process (i.e. renal failure or EtOH abuse) that can also explain the patient's hyponatremia.

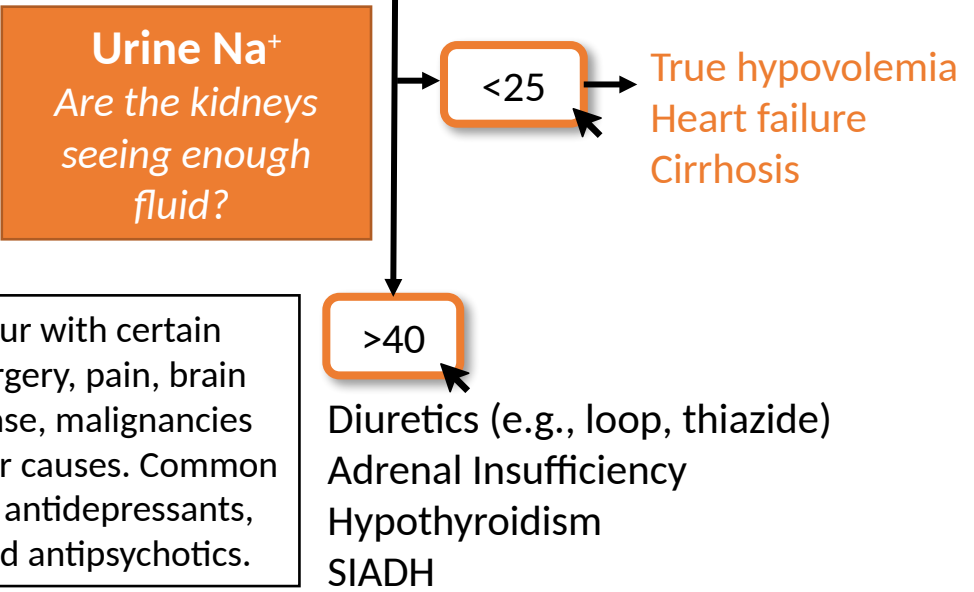
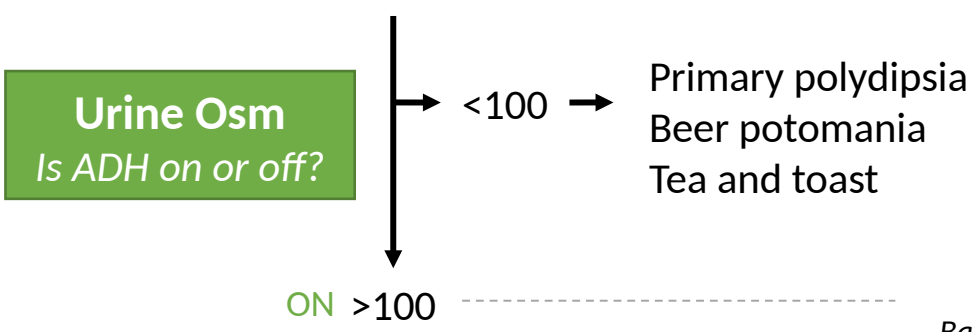
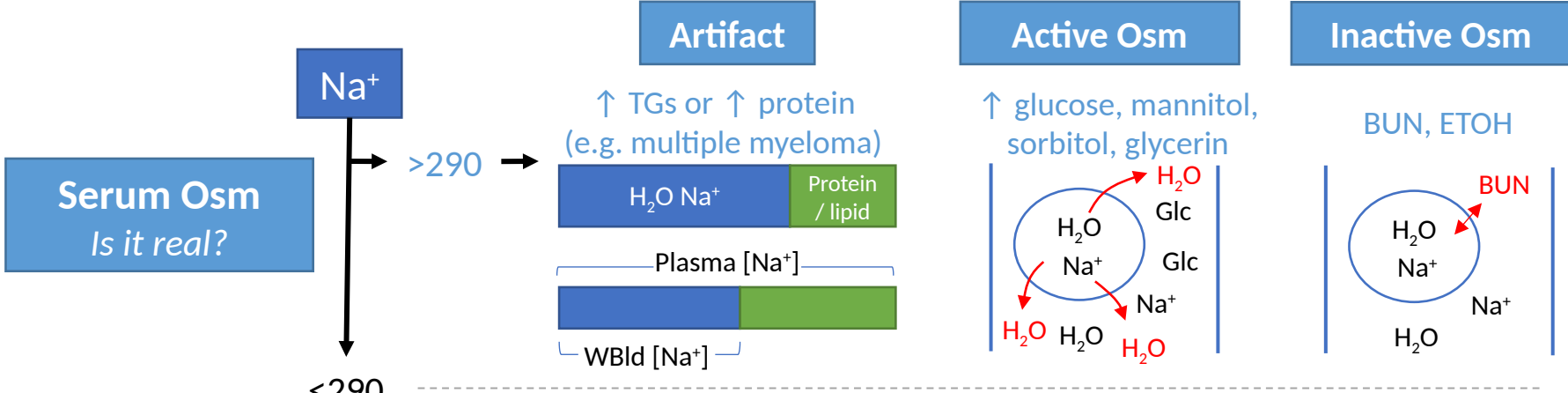
- Step 1: S_{Osm}
- Step 2: U_{Osm}
- Step 3: U_{Na}
- Cases



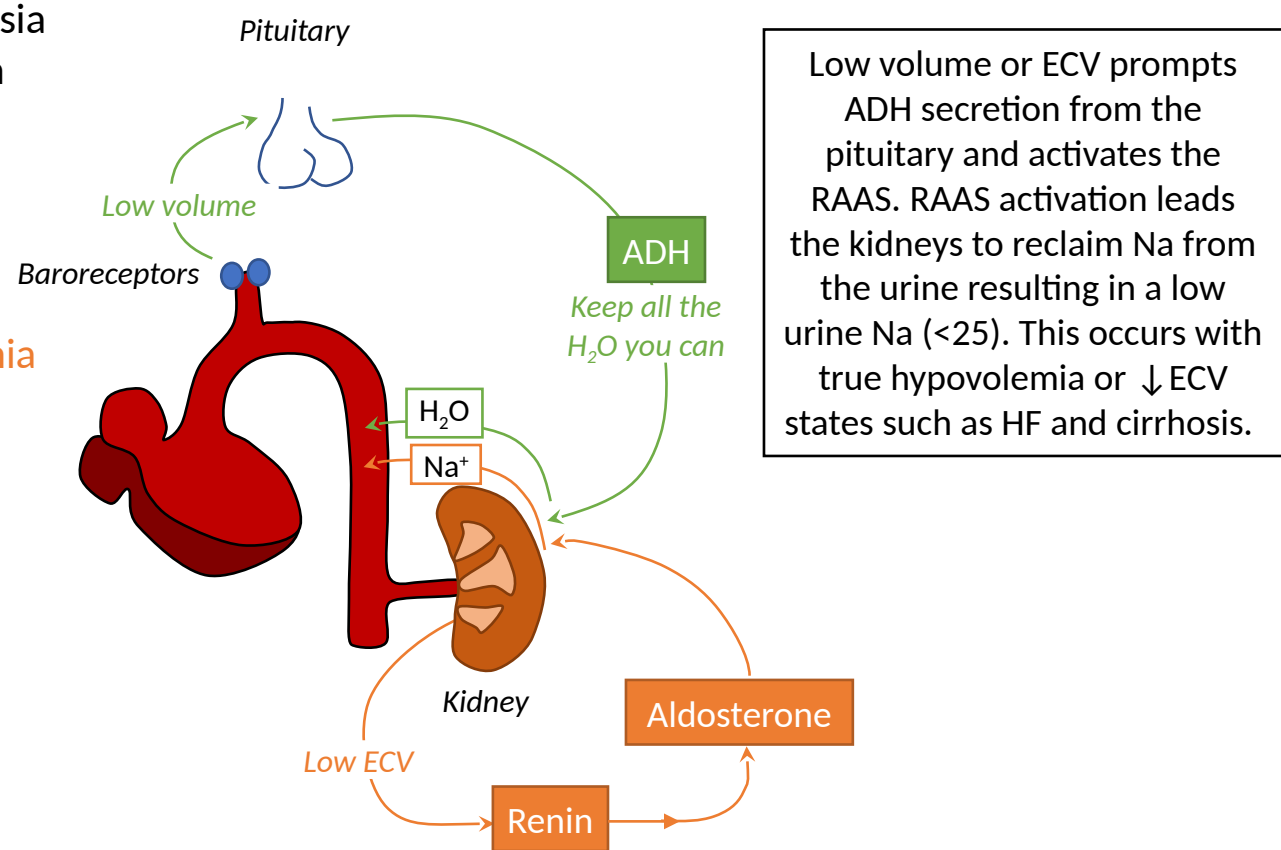
Most cases of hypo-osmolar hyponatremia result in high ADH state. The next differentiating question in these cases is what the trigger for ADH secretion and whether ADH secretion is appropriate or not.



- Step 1: S_{Osm}
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SIADH can occur with certain medications, surgery, pain, brain injury, lung disease, malignancies among many other causes. Common drug culprits are antidepressants, antiepileptics, and antipsychotics.



Step 1: S_{Osm}

Step 2: U_{Osm}

Step 3: U_{Na}

Cases

Case 1

40 yo F with nausea/vomiting and diarrhea. Serum Na is 125.

1. Serum osm 265
2. Urine osm 400
3. Urine Na < 10

Serum osm is low, or true hypoNa

Urine osm $\uparrow$ suggesting $\uparrow$ ADH state.

Urine Na is low suggesting low ECV state.

Given clinical history of nausea/vomiting and diarrhea, hyponatremia is likely from true hypovolemia.

Case 2

21 yo M with type I DM who presents with nausea/vomiting and abdominal pain. He presents with blood glucose of 500 and + ketones on UA. Serum Na is 125.

1. Serum osm 310
2. Urine osm 800
3. Urine Na 20

Hyperosm hypoNa. "something else in the blood." Glc is the active osm.
corrected Na = measured Na + $[1.6 (\text{glucose} - 100) / 100]$
The calculated Na is 131.

Urine osm $\uparrow$ suggesting $\uparrow$ ADH state.

Urine Na is borderline-low suggesting low ECV.

The patient has hyperosmolar hypoNa 2/2 to excess Glc. However, they also have true hypoNa (that is mild at 131). This hyponatremia is 2/2 to hypovolemia.

Case 3

68 yo M with a history of alcohol use disorder presents with alcohol withdrawal. Serum Na is 125.

1. Serum osm 270
2. Urine osm 60
3. Urine Na 65

Serum osm is low, or true hypoNa.

Urine osm is normal, suggesting ADH is off. This only occurs in poor solute intake or excess water intake.

Not needed in this case.

Given alcohol use history, this is likely beer potomania